



MGCNet: Multiple group-wise correlation network with hierarchical contrastive learning for co-salient object detection

Xian Fang^a, Xin Wang^a, Jinchao Zhu^{b,c},*, Qiaohong Chen^a, Zuofan Chen^a

^a School of Computer Science and Technology, Zhejiang Sci-Tech University, Hangzhou 310018, China

^b College of Software, Nankai University, Tianjin 300350, China

^c Department of Automation, Tsinghua University, Beijing 100089, China

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ABSTRACT

Co-salient object detection (CoSOD) aims to discover common salient objects from a group of relevant images. Existing methods typically extract inter-image correspondences in one or two image groups, which could limit the ability of model to learn high-quality representations. To overcome this limitation, we propose a novel multiple group-wise correlation network (MGCNet), which thoroughly learns inter-image relations by simultaneously processing positive and negative samples among four image groups. Initially, to enhance the discriminative capability of model, we introduce the multi-level contrastive learning module (MCLM), which pulls positive sample pairs closer and pushes negative sample pairs farther at both the global and local levels by applying contrastive learning. Subsequently, to enhance consensus features more comprehensively, we propose the dual enhancement module (DEM), which sufficiently explores both the channel and spatial dimensions in a two-stage manner. Finally, to retain the integrity of the consensus features, we design the consensus retaining module (CRM), which effectively fuses neighboring level features to complement the location information within the consensus features. Extensive experiments on three challenging benchmark datasets demonstrate that our MGCNet achieves excellent performance compared to 17 state-of-the-art models.

1. Introduction

Co-salient object detection (CoSOD) aims at detecting common salient objects from image groups containing two or more related images. Compared to the standard salient object detection (SOD) task, CoSOD is more challenging, and requires identifying shared objects from multiple salient objects while eliminating the interference of non-salient objects appearing in different images. The CoSOD approaches have also proved their advantages in the preprocessing stage of many visual tasks, such as object co-segmentation [1,2], video colocalization [3,4], and weakly supervised semantic segmentation [5,6].

Most existing work tends to accomplish the CoSOD task by exploring shared features [7] and common semantic information [8] within a single image group to obtain various coherence information between images. Although these approaches achieve favorable performance in many scenarios, most of them rely solely on a single image group to obtain the relations between images, which may cause some limitations. First, the number of images in a single image group is usually insufficient to mine semantic clues, often resulting in the model generating

unsatisfactory predictions that include salient objects from other categories. Second, a single image group may struggle to provide sufficient negative relations, which could prevent most models from accurately distinguishing different kinds of co-salient objects. To alleviate these problems, some recent works [9,10] propose models based on two image groups, which compensates for the shortcomings of a single image group being unable to form discriminative representation to some extent. However, due to the large amount of object categories and contextual information contained in the images of CoSOD task, exploring two image groups may not be adequate for model to capture the diversity of complementary features from co-salient objects, which limits the ability of CoSOD model to fully exploit discriminative feature representations.

Recently, contrastive learning has attracted a great deal of attention in the field of self-supervision, even outperforming supervised learning methods on some specific computer vision tasks. Contrastive learning derives discriminative representations by modeling between similar and dissimilar pairs, pulling similar samples closer and pushing dissimilar ones farther in the feature space, which naturally inspires us to design suitable contrastive learning methods to solve the CoSOD

* Corresponding author.

E-mail address: jczhu@nankai.edu.cn (J. Zhu).

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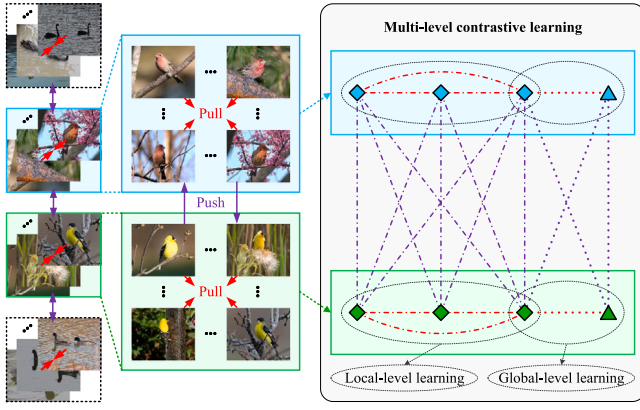


Fig. 1. Main idea of our proposed multi-level contrastive learning. The method effectively optimizes the distance between similar and dissimilar samples in the feature space by applying contrastive learning at both the global and local levels among multiple image groups.

problem. In recent years, a number of CoSOD methods [9–11] have attempted to use contrastive learning or alternative approaches to help models learn discriminative consensus features, achieving promising performance. However, these methods are all restricted to single or two image groups to construct positive and negative sample pairs, which limits the effectiveness of contrastive learning to a large extent. Therefore, constructing more positive and negative sample pairs borrowing contrastive learning is crucial for refining the model.

In this paper, we optimize the distance between samples by pulling samples from the same group closer and pushing samples from different groups farther with contrastive learning. As shown in Fig. 1, we propose multi-level contrastive learning to minimize the distance within groups and maximize the distance between groups. Based on this method, we bring forward a multiple group-wise correlation network (MGCNet) that focuses on richer potential relations between four image groups. Our MGCNet efficiently extracts intra-group consensus features and distinguishes inter-group object attributes by exploring positive and negative relations among four image groups without using additional co-classification information. Specifically, we first design a multi-level contrastive learning module (MCLM) by forming more diverse positive and negative relations from the inter-group images at both global and local levels, avoiding the introduction of additional co-classification information, which enables the model to better distinguish co-salient objects from different image groups. Next, to further mine the intra-group consensus information, we design a dual-enhancement module (DEM), which can comprehensively extract the intra-group consensus features from both the spatial and channel dimensions. Finally, since the extracted consensus information will be gradually diluted during the process of propagating to the low-level layer, we design the consensus retaining module (CRM) to fuse the features of two neighboring layers from the encoder and add them to the decoder.

Our main contributions can be summarized as follows:

- We propose a novel multiple group-wise correlation network (MGCNet), which obtains more positive and negative relations from four image groups to identify the inherent connections between images. Additionally, we propose the multi-level contrastive learning module (MCLM) to pull the positive samples closer and push the negative samples farther at both global and local levels.
- We design the dual enhancement module (DEM) to comprehensively aggregate and enhance the consensus features from both spatial and channel dimensions in a two-stage manner. Meanwhile, we introduce the consensus retaining module (CRM) to supplement the DEM by preventing the gradual dilution of consensus features during propagation.
- We conduct extensive experiments on three challenging datasets, including Cosal2015, CoSOD3k, and CoCA, verifying the effectiveness of MGCNet and outperforming most existing models.

2. Related work

2.1. Salient object detection

The purpose of salient object detection (SOD) is to segment the objects or regions of an image that attract the most attention. Early work on SOD focused on a series of explorations around handcrafted features [12–15]. These methods typically leverage known cues to design classification models that extract features from object candidate boxes [16–18] or super-pixels [19–22]. Although these methods have improved in accuracy, they require considerable time to extract target regions and features. Nowadays, with the successful application of fully convolutional networks [23] in the field of salient object detection, most of the researches mainly pay attention to effectively fuse high-level and low-level features based on the end-to-end UNet [24] architecture. For instance, PFAN [25] enhanced high-level features and low-level features separately and fused them sufficiently to exploit their different effects. CPD [26] designed a partial decoder to improve the detection performance by discarding the low-level features and mining the deeper ones. TCRNet [27] proposed to aggregate global, deep, and shallow information in a cascaded manner while suppressing background noise. PoolNet+ [28] integrated features from different levels and increases the proportion of pooling operations to reduce computational costs. Some works employ attention mechanisms to further enhance features. PAGRN [29] proposed a progressive attention-driven framework, using multi-path recurrent feedback to enhance and refine the network. PiCANet [30] concentrated on the use of contextual attention at the pixel level to selectively aggregate contextual information. LLISOD [31] enhances the detection performance of the model in low-light conditions by hierarchical feature polishing strategy. In addition, some works introduce additional information to guide the model. BASNet [32] proposed a predict-refine architecture to continuously optimize the saliency maps by focusing on the quality of boundaries. EGNNet [33] facilitated the model to locate salient objects more accurately by extracting salient object features and salient edge features separately and coupling them.

2.2. Co-salient object detection

In contrast to traditional SOD, CoSOD emphasizes on locating co-salient objects in a set of semantically related images rather than in a single image. Early CoSOD methods generally utilized handcrafted cues like intensity, edge, attention cues, color and texture [34] to explore the low-level consistency [35] of the pictures within a group of images, such as clustering [7], self-adaptive weighting [36], and metric learning [8]. However, these methods struggle to capture complex co-salient information, often resulting in poor model performance. At present, with the continuous development of deep learning, many end-to-end deep CoSOD models have been proposed. CSMG [37] proposed a hierarchical image co-saliency detection framework that guides the model by extracting masks and designs multi-scale label smoothing model to optimize the co-saliency map. CoEGNet [38] obtained co-saliency map by combining the saliency map acquired from salient object detection model with the co-saliency map gained from co-saliency projection. GCAGC [39] explored pixel-level coherence by modeling intra-image and inter-image relations using graph convolutional networks. ICNet [40] preferred to utilize the intra-saliency cues to improve CoSOD performance. GICD [41] and DeepAGC [42] structured gradient information and Gromov–Wasserstein distance to measure pixel similarity, inducing the model to pay more attention to the co-salient objects. GLNet [43] proposed a global-and-local collaborative learning architecture to collaboratively explore connections between images from both global and local perspectives. GCoNet [9] and GCoNet+ [10] explore inter-group separability and intra-group compactness by utilizing two image groups at the group level. CoAD-Net [44] collaboratively extracted group semantic information and

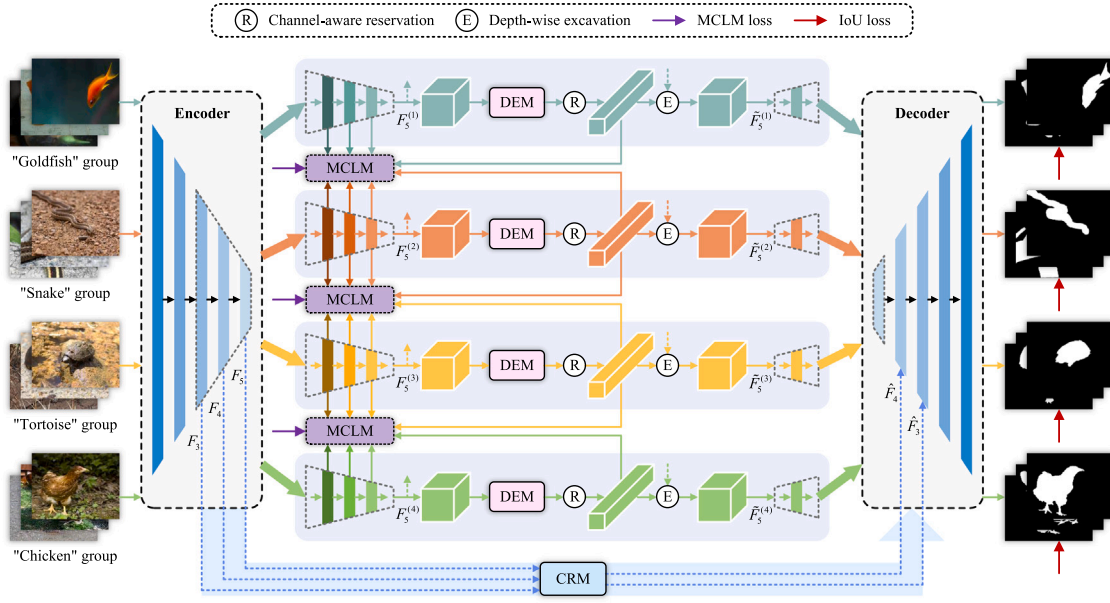


Fig. 2. The architecture of the proposed MGCNet. The network consists of three main components, including multi-level contrastive learning module (MCLM), dual enhancement module (DEM), and consensus retaining module (CRM). Note that the MCLM is only adopted during training.

generated co-salient features through group-attentional semantic aggregation and gated group distribution. CoRP [45] continuously purified the co-representation in an iterative manner to minimize the effect of noise. CADC [46] and CADC++ [47] adopted dynamic convolution to extract consensus features and locate co-salient objects. TCNet [48] captured global and local dependent relations via parallel interaction of Transformers and CNNs. DMT [49] designed a two-branch structure to mine co-saliency information while exploring the influence of background information on feature extraction. However, these methods all extract co-salient information from one or two image groups, which could lead to their prediction maps being easily interfered by other irrelevant objects.

2.3. Contrastive learning

Most recently, contrastive learning has rapidly developed as a popular discriminative method for representation learning in self-supervised classification and has achieved encouraging performance in many other tasks, such as semantic segmentation [50], language pre-training [51], and visual localization [52]. These methods treat augmented versions of the same instance as positive pairs and instances from different categories as negative pairs, learning representations through the comparison between positive and negative pairs [53]. Recently, SimCLR [54] investigated that increasing the number of positive and negative samples is beneficial for contrastive learning. MoCo [55] allowed the model to express richer visual features by building a sufficiently large and feature-consistent dynamic dictionary. SCL [56] proposed supervised contrastive learning for image classification. This approach enables the model to learn more discriminative feature representations by designing a contrastive loss that can handle multiple pairs of positive and negative samples. In the field of SOD, TPCL [57] proposed the application of contrastive learning method at the pixel-level, enabling the model to obtain strictly discriminative pixel-level feature representations. ICONwC [58] explored contrastive learning at the super-pixel level to sample more diverse and representative pixels. For the CoSOD task, DCFM [11] generated positive and negative sample pairs from the image itself by designing a self-contrastive learning approach. Although the method achieves favorable performance improvement, achieving effective training results is challenging for contrastive learning when positive and negative samples are derived from only a single image

Table 1
Some important notations used in this paper.

Notations	Descriptions
i	Group index
j	Layer index of encoder and decoder
G_i	Input image group
F_j	Output feature of encoder
$F_j^{(i)}$	Output feature of encoder in different image groups
$\hat{F}_j^{(i)}$	Consensus feature
$\hat{F}_j$	Middle-level input feature of decoder
$\hat{F}_5^{(i)}$	High-level input feature of decoder

group. Therefore, we propose to use contrastive learning at both the global and local levels, which can process the features obtained from four image groups more finely and help the model better distinguish different object attributes.

3. Methodology

3.1. Overview

The overall architecture of the proposed MGCNet is illustrated in Fig. 2. Initially, our model simultaneously incorporates four groups of source images as inputs and extracts features through an encoder. Subsequently, we design the dual enhancement module (DEM) to effectively extract consensus features from the group. Furthermore, we design the consensus retaining module (CRM) to complement consensus features in multiple layers, ensuring the integrity of co-salient information. After that, the consensus features of different groups are distinguished by the proposed multi-level contrastive learning module (MCLM). Finally, the generated features of each group are input to the decoder to obtain the final prediction map.

We provide some important notations adopted in this section in Table 1. Specifically, four image groups G_i ($i = 1, 2, \dots, 4$) containing four types of co-salient objects are given as inputs, where i stands for the i th group. Initially, these images are fed into an encoder of VGG-16 [59], to extract backbone features. We retain three high-level features for each group of encoder. For simplicity, we denote these features as $\{F_3^{(i)}, F_4^{(i)}, F_5^{(i)}\} \in \mathbb{R}^{N \times C \times H \times W}$, where N represents the size of the group, C represents the number of channels, and $H \times W$ is the size

of the space. Next, each group of high-level features $F_5^{(i)}$ are separately fed into the DEM, which aggregate features from spatial and channel dimensions in a two-stage manner. Then, channel-aware reservation is applied to generate the consensus features $\bar{F}^{(i)} \in \mathbb{R}^{1 \times C \times 1 \times 1}$. Meanwhile, we introduce CRM to supplement the consensus features that may be diluted during the propagation process. This module fuses three high-level features $\{F_3, F_4, F_5\}$ from the encoder and delivers the fused features $\{\hat{F}_3, \hat{F}_4\}$ to the third and fourth layers of the decoder. Besides, to more accurately distinguish the features of different groups, we input the high-level features $\{F_3^{(i)}, F_4^{(i)}, F_5^{(i)}\}$ and the consensus feature $\bar{F}^{(i)}$ between two neighboring groups into the multi-level contrastive learning module. This module adopts the contrastive learning method at the global and local levels and is supervised by the designed MCLM loss to better explore different group consensus cues among them. Finally, the generated consensus features of each group are depth-wise excavated with the original features $F_5^{(i)}$ to generate the enhanced features $\tilde{F}_5^{(i)} \in \mathbb{R}^{N \times C \times H \times W}$ and fed it into the decoder to obtain the final co-salient object prediction map, which is supervised by IoU loss.

3.2. Multi-level contrastive learning module

Currently, most of the CoSOD methods concentrate on detecting co-salient objects within a single image group, while ignoring the ability to discriminate between different co-salient objects from multiple image groups. This situation may lead to the model incorrectly detecting non co-salient objects when recognizing an image containing multiple salient objects. Although a few existing methods attempt to mine the discriminative properties between different image groups using techniques such as metric learning, the representations learned from only a limited number of samples are insufficient for the complex CoSOD task. For this reason, we propose the multi-level contrastive learning module (MCLM) to strengthen the ability of the model in capturing and distinguishing rich contextual information among the four image groups simultaneously.

The structure of MCLM is illustrated in Fig. 3. Note that our MCLM is only activated during the training process and will be removed during the inference stage. Since we have applied the MCLM among multiple groups, we will adopt the 1st and 2nd groups as examples to illustrate the MCLM for simplicity. Given the inputs $\{F_j^{(i)}, \bar{F}^{(i)}\}$ ($i \in \{1, 2\}$, $j \in \{3, 4, 5\}$), we first perform channel-aware reservation for $F_j^{(i)}$ to acquire features $\tilde{F}_j^{(i)} \in \mathbb{R}^{1 \times C \times 1 \times 1}$. After that, we feed $\{\tilde{F}_j^{(i)}, \bar{F}^{(i)}\}$ into a projection layer, each consisting of a sequential combination of MLP and ReLU repeated twice, to generate the features $\{\tilde{F}_j^{(i)}, \bar{F}^{(i)}\}$. Since $\bar{F}^{(i)}$ refers to the features with global information extracted through transformer block existing in DEM, we adopt a hierarchical design for the contrastive learning. This hierarchical design allows the model to progressively refine feature distinctions at different scales, ensuring a finer and more accurate separation of relevant and irrelevant information throughout the learning process. Specifically, we regard $\tilde{F}_j^{(i)}$ as local-level samples and $\bar{F}^{(i)}$ as global-level samples. Then we construct positive pairs of samples between features in the same group and negative pairs of samples between features in different groups. Simultaneously, a MCLM loss is designed to pull positive sample pairs closer and push negative sample pairs farther. Initially, we introduce the cosine-style similarity between the sample pairs by:

$$\cos(F_1, F_2) = (1 + \frac{F_1 \cdot F_2}{|F_1| \cdot |F_2|}) \times 0.2, \quad (1)$$

where $|\cdot|$ represents the L_2 norm. This procedure evaluates the relative position and similarity of two vectors in feature space by measuring their similarity in direction. Following this, we define the local-level MCLM loss $\mathcal{L}_l^{(1,2)}$ between the 1st and 2nd groups, which can be formalized as:

$$\mathcal{L}_l^{(1,2)} = \mathcal{L}_{lp}^{(1,2)} + \mathcal{L}_{ln}^{(1,2)}, \quad (2)$$

Table 2

The detailed values of F_1 and F_2 in $\cos(F_1, F_2)$.

Loss term	Corresponding values		
$\mathcal{L}_{lp}^{(1,2)}$	$(\tilde{F}_3^{(1)}, \tilde{F}_4^{(1)})$ $(\tilde{F}_3^{(2)}, \tilde{F}_4^{(2)})$	$(\tilde{F}_3^{(1)}, \tilde{F}_5^{(1)})$ $(\tilde{F}_3^{(2)}, \tilde{F}_5^{(2)})$	$(\tilde{F}_4^{(1)}, \tilde{F}_5^{(1)})$ $(\tilde{F}_4^{(2)}, \tilde{F}_5^{(2)})$
$\mathcal{L}_{ln}^{(1,2)}$	$(\tilde{F}_3^{(1)}, \tilde{F}_5^{(2)})$ $(\tilde{F}_4^{(1)}, \tilde{F}_5^{(2)})$ $(\tilde{F}_5^{(1)}, \tilde{F}_3^{(2)})$ $(\tilde{F}_5^{(1)}, \tilde{F}_4^{(2)})$	$(\tilde{F}_3^{(1)}, \tilde{F}_5^{(2)})$ $(\tilde{F}_4^{(1)}, \tilde{F}_5^{(2)})$ $(\tilde{F}_5^{(1)}, \tilde{F}_3^{(2)})$ $(\tilde{F}_5^{(1)}, \tilde{F}_4^{(2)})$	$(\tilde{F}_3^{(1)}, \tilde{F}_5^{(2)})$ $(\tilde{F}_4^{(1)}, \tilde{F}_5^{(2)})$ $(\tilde{F}_5^{(1)}, \tilde{F}_3^{(2)})$ $(\tilde{F}_5^{(1)}, \tilde{F}_4^{(2)})$
$\mathcal{L}_{gp}^{(1,2)}$	$(\tilde{F}_5^{(1)}, \bar{F}^{(1)})$	$(\tilde{F}_5^{(2)}, \bar{F}^{(2)})$	
$\mathcal{L}_{gn}^{(1,2)}$	$(\tilde{F}_5^{(1)}, \bar{F}^{(2)})$	$(\tilde{F}_5^{(2)}, \bar{F}^{(1)})$	$(\bar{F}^{(1)}, \bar{F}^{(2)})$

where the loss $\mathcal{L}_l^{(1,2)}$ is composed of the local-level positive loss $\mathcal{L}_{lp}^{(1,2)}$ and the local-level negative loss $\mathcal{L}_{ln}^{(1,2)}$, which can be respectively defined as:

$$\begin{aligned} \mathcal{L}_{lp}^{(1,2)} &= - \sum_{i=1}^2 \sum_{j=3}^4 \sum_{k=j+1}^5 \log(\cos(\tilde{F}_j^{(i)}, \tilde{F}_k^{(i)}) + \theta), \\ \mathcal{L}_{ln}^{(1,2)} &= - \sum_{j=3}^5 \sum_{k=3}^5 \log(1 - \cos(\tilde{F}_j^{(1)}, \tilde{F}_k^{(2)}) + \theta), \end{aligned} \quad (3)$$

where θ is set as 1×10^{-5} to ensure non-zero values for $\log(\cdot)$, and the global-level MCLM loss $\mathcal{L}_g^{(1,2)}$ between the 1st and 2nd groups can be formalized as:

$$\mathcal{L}_g^{(1,2)} = \mathcal{L}_{gp}^{(1,2)} + \mathcal{L}_{gn}^{(1,2)}, \quad (4)$$

where the loss $\mathcal{L}_g^{(1,2)}$ is composed of the global-level positive loss $\mathcal{L}_{gp}^{(1,2)}$ and the global-level negative loss $\mathcal{L}_{gn}^{(1,2)}$, which can be respectively defined as:

$$\begin{aligned} \mathcal{L}_{gp}^{(1,2)} &= - \sum_{i=1}^2 \log(\cos(\tilde{F}_5^{(i)}, \bar{F}^{(i)}) + \theta), \\ \mathcal{L}_{gn}^{(1,2)} &= - \sum_{i=1}^2 \sum_{j=1}^2 \log(1 - \cos(\tilde{F}_5^{(i)}, \bar{F}^{(j)}) + \theta) + t, i \neq j, \end{aligned} \quad (5)$$

where

$$t = -\log(1 - \cos(\bar{F}^{(1)}, \bar{F}^{(2)}) + \theta). \quad (6)$$

The detailed values of F_1 and F_2 in $\cos(F_1, F_2)$ are shown in Table 2. Considering that the process of calculating $\mathcal{L}_l^{(2,3)}$ and $\mathcal{L}_l^{(3,4)}$ is similar that of $\mathcal{L}_l^{(1,2)}$, and the process of calculating $\mathcal{L}_g^{(2,3)}$ and $\mathcal{L}_g^{(3,4)}$ is also similar that of $\mathcal{L}_g^{(1,2)}$, we do not repeat their process here. Eventually, the MCLM loss across four groups is defined as:

$$\mathcal{L}_{MCLM} = \sum_{i=1}^3 (\mathcal{L}_l^{(i,i+1)} + \mathcal{L}_g^{(i,i+1)}). \quad (7)$$

The proposed MCLM enables the model to learn a more discriminative consensus by establishing contrastive learning across multiple relations at both global and local levels. This will make the network better at excluding interference from other salient objects in complex environments.

3.3. Dual enhancement module

Objects of the same class in real life usually have common or similar appearance. Some approaches seek such consensus features only from a one-sided perspective, often failing to identify consensus features that are truly representative of the image group. The proposed dual enhancement module (DEM) captures not only the broad spatial context but also the fine-grained channel details in a two-stage manner, which provides more complete and precise representations of co-salient objects.

The DEM structure is illustrated in Fig. 4. We design a two-stage aggregation framework to explore correlation information within groups.

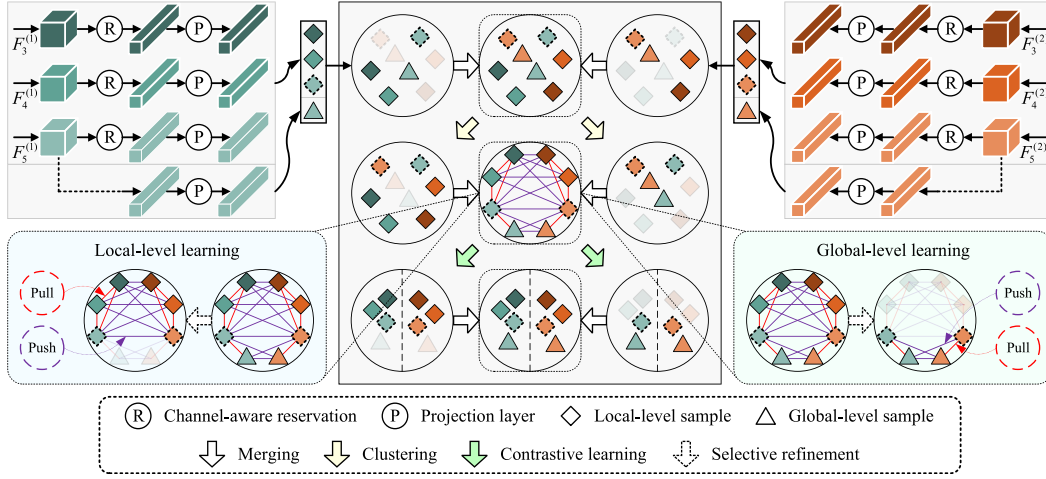


Fig. 3. The illustration of multi-level contrastive learning module (MCLM). Firstly, the three high-level features of encoder $F_j^{(i)}$ and the enhanced feature of $F_5^{(i)}$ from the 1st and 2nd groups are input into the MCLM and processed group by group. Subsequently, the processed features are clustered for selective refinement. Finally, the distance between different samples is optimized by local-level and global-level contrastive learning.

The first stage focuses on splitting and processing channel information, while the second stage concentrates on the fusion of spatial information at different scales. For simplicity, we take the 1st group as an example to illustrate the DEM.

In the 1st stage, we introduce feature recombination to aggregate features in the channel dimension. Inspired by channel shuffling [60], we first apply channel attention to the input feature $F_5^{(1)}$ and separate it into four streams along the channel dimension to obtain $C_i \in \mathbb{R}^{N \times C/4 \times H \times W}$. Subsequently, to achieve deeper channel mixing, we further divide each group of features into four equal parts along the channel dimension to produce the features $C_{ij} \in \mathbb{R}^{N \times C/16 \times H \times W}$, where ij represents the j th feature of the i th group. Afterwards, to facilitate more effective feature fusion, we then recombine the features of each group. Specifically, we take the j th feature from C_{ij} to generate the recombined features C_j respectively. For instance, we take $\{C_{11}, C_{21}, C_{31}, C_{41}\}$ to generate the recombined feature C_1 . Finally, we compute the average value of the recombined feature along the channel dimension and correlate it with the original feature. The formula can be expressed as:

$$\begin{aligned} C_i &= FR(FS(FS(CA(F_5^{(1)}))))), i = 1, 2, \dots, 4, \\ C &= Cat(C_1, C_2, C_3, C_4), \\ C' &= (\sigma(Conv(\frac{1}{N} \sum_{n=1}^N C[\cdot, n, \cdot, \cdot])) \odot F_5^{(1)}) \oplus F_5^{(1)}, \end{aligned} \quad (8)$$

where CA represents the channel attention, FS and FR denote the feature split and recombination operation, $Conv$ is the convolutional layer of size 3×3 , σ is the sigmoid function, Cat denotes concatenation operation, $\odot$ means element-wise multiplication, and $\oplus$ means element-wise summation. In this way, we thoroughly mix features from different channels, using the attention mechanism to emphasize significant features while suppressing irrelevant ones. The convolutional layers of the encoder may not be sufficiently sensitive to the spatial locations of the features during extraction, whereas the vision transformer [61] is capable of capturing the global positional information of the input features. Therefore, for the input feature C' , we adopt a transformer block to enhance the representation of global co-salient objects. Concretely, we reshape C' into $T \in \mathbb{R}^{N \times (H \times W) \times C}$, and then input T into the transformer block to generate the enhanced feature T'' , the formula for this process is expressed as:

$$\begin{aligned} T' &= (LN(MHSA(T))) \oplus T, \\ T'' &= (LN(MLP(T'))) \oplus T', \end{aligned} \quad (9)$$

where LN stands for the layernorm [62] function, $MHSA$ represents the multi-head self-attention, and MLP means the multi-layer perceptron. After that, we reshape T'' back into the feature $F_5^{(1)'} \in \mathbb{R}^{N \times C \times H \times W}$.

In the 2nd stage, we aggregate features at the spatial level. We observe that applying both average pooling and maximum pooling for multiple receptive fields can significantly enhance the ability of model to capture contextual information from multiple regions. Therefore, we first adopt two different sizes of adaptive average pooling and adaptive maximum pooling for the input feature $F_5^{(1)'}$. Then, we apply a convolutional block to the pooled features and fuse the features obtained from the same receptive fields. Next, we fuse the obtained features again and correlate the fused feature with the original feature. This process can be expressed as:

$$\begin{aligned} S_n &= CBR(AAP_n(F_5^{(1)'})) \oplus CBR(AMP_n(F_5^{(1)'})), n = 1, 5, \\ S' &= (CBR(\sigma(S_1) \oplus DW(\sigma(S_5))) \odot F_5^{(1)'}) \oplus F_5^{(1)'}, \end{aligned} \quad (10)$$

where CBR stands for the combination of convolution layer, batch normalization and ReLU, AAP_n and AMP_n denote adaptive average pooling and adaptive maximum pooling with an output size of $n \times n$, and DW represents the downsampling operation. As can be seen, we ensure that the model captures not only global but also local contextual information, which is crucial for detecting finer details of co-salient objects. Finally, we input feature S' into the same transformer block as in the first stage to get the final feature $F_5^{(1)''}$.

The DEM effectively captures the shared attributes of co-salient objects within an image group by deeply mining the channel and spatial information of the images, thereby remarkably reinforcing the intrinsic connections and consistency among the images within the group.

3.4. Consensus retaining module

We build our network based on the feature pyramid network (FPN) [63]. Although FPN performs well in multi-scale object detection, the propagation of the deep features from the deep layer to the shallow layer results in the gradual deficiency of position information due to the interference of low layer information. In addition, it is well known that the low-level features of the encoder primarily contain the detailed information, while the high-level features encapsulate the semantic information, so the fusion of features at different levels can leverage their complementary advantages. Based on the reasons above, we design the consensus retaining module (CRM), which retains the integrity of co-salient information by progressively fusing the information from different levels of encoder and supplementing it to the corresponding level of decoder.

The structure of CRM is illustrated in Fig. 5. Since the low-level features offer minimal improvement to the model performance while huge computational cost [26], we only utilize the high-level features

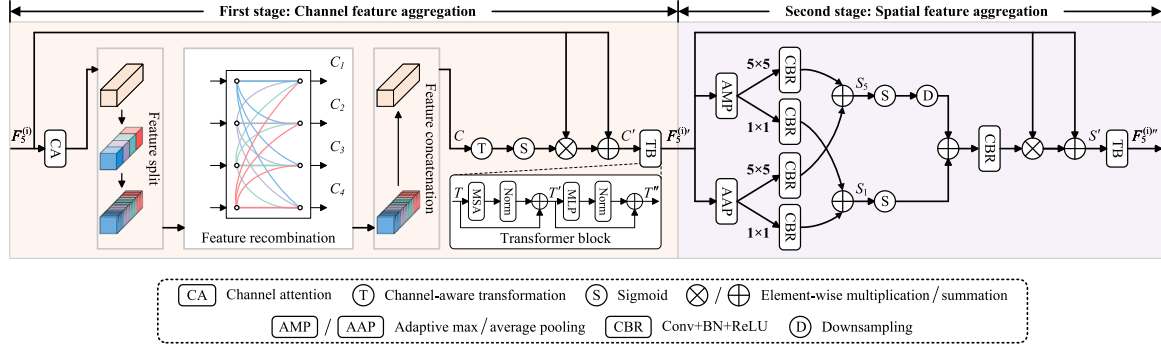


Fig. 4. The illustration of dual enhancement module (DEM). In the first stage, the input feature $F_5^{(i)}$ aggregates channel information mainly through feature split, recombination and concatenation. In the second stage, the aggregated feature $F_5^{(j)}$ is further mined to extract spatial information and generate output feature $F_5^{(j)'}$ through adaptive max and average pooling at different scales.

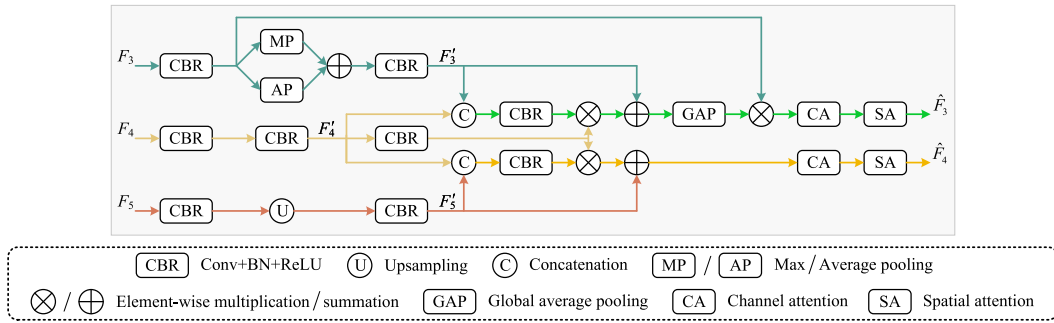


Fig. 5. The illustration of consensus retaining module (CRM). Different levels of input features $\{F_3, F_4, F_5\}$ are progressively fused and enhanced to generate output features $\{\hat{F}_3, \hat{F}_4\}$.

from the last three layers of encoder, which have lower resolution and computational cost. Given the input features $\{F_3, F_4, F_5\}$, we first utilize a convolutional block for each of these features individually, which allows us to refine their local information and improve feature specificity. Then, to align high-level and low-level features at different scales, we upsample the high-level features and apply both maximum pooling and average pooling to downsample the low-level features. Subsequently, an additional convolutional block is applied to each of the three feature branches, further enhancing their representational capacity in detail. This process can be formulated as:

$$\begin{aligned} F_3' &= CBR(AP(CBR(F_3)) \oplus MP(CBR(F_3))), \\ F_4' &= CBR(CBR(F_4)), \\ F_5' &= CBR(UP(CBR(F_5))), \end{aligned} \quad (11)$$

where AP and MP represent average pooling and max pooling respectively, and UP denotes the upsampling operation. Subsequently, we perform channel-wise concatenation on features from adjacent layers to combine multi-level information, and then feed them into a convolutional block to facilitate the initial integration of local and global feature representations. Next, to achieve deep fusion between neighboring branches, we correlate the obtained features with the middle-level features respectively. Finally, we resize the spatial dimensions of the lower-level branch to match the features in the decoder and apply channel attention and spatial attention to the features of two branches. This dual attention mechanism effectively highlights co-salient regions, ensuring the network focuses on the most relevant areas across the input images. The specific formulas are provided as follows:

$$\begin{aligned} \hat{F}_3 &= CS(G((CBR(Cat(F_3', F_4')) \odot CBR(F_4')) \oplus F_3') \odot F_3^c), \\ \hat{F}_4 &= CS((CBR(Cat(F_4', F_5')) \odot CBR(F_4')) \oplus F_5'), \end{aligned} \quad (12)$$

where CS means the combination of channel attention and spatial attention, and G represents global average pooling, F_3^c denotes F_3 with a convolutional block.

In summary, CRM facilitates mutual fusion of features from three neighboring levels. It makes full use of the information at different levels by deeply mining the detailed information of low-level features and the rich semantic information of high-level features, thereby achieving the interactive fusion of this information.

3.5. Loss function

In addition to our MCLM loss, we also employ Intersection over Union (IoU) loss [32,44] to supervise the saliency maps, which optimizes the accuracy of the segmentation region in the prediction map. The IoU loss can be illustrated as:

$$\mathcal{L}_{iou} = 1 - \frac{1}{N} \sum_{i=1}^N \frac{\hat{Y}_i \cap Y_i}{\hat{Y}_i \cup Y_i}, \quad (13)$$

where $\hat{Y}$ and Y are the prediction and groundtruth, and N means the number of images.

The final loss function is a combination of our MCLM loss and IoU loss. This combined loss function leverage the strengths of both components, ensuring that our model not only distinguishes objects in different groups effectively but also produces highly precise segmentation maps. The final loss function is denoted as:

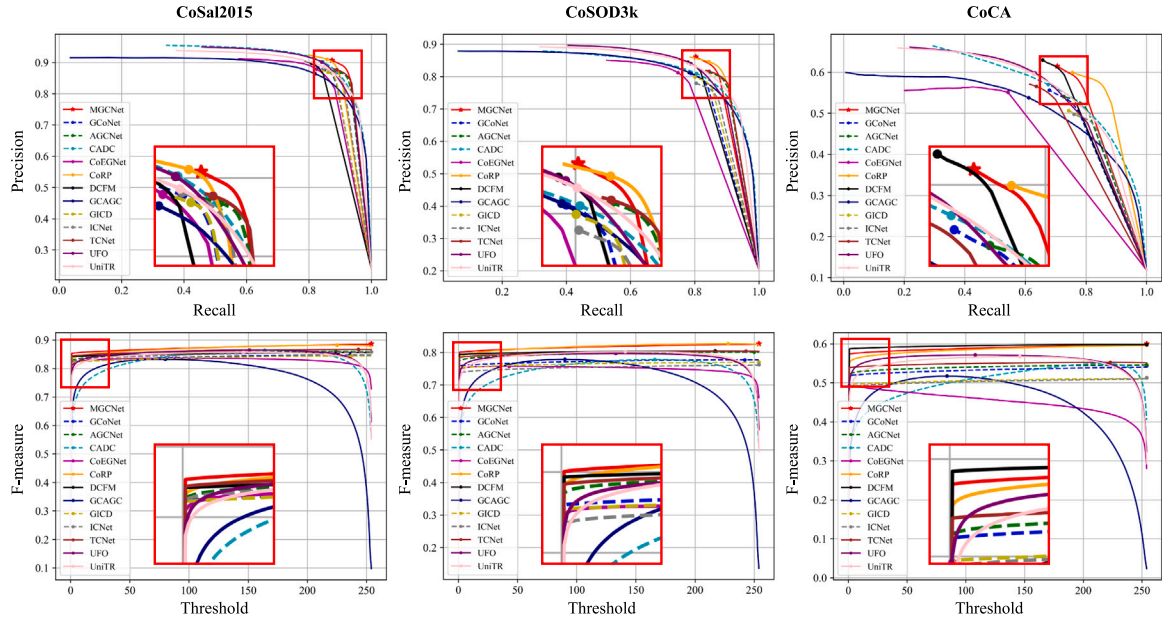
$$\mathcal{L}_{total} = \lambda \mathcal{L}_{MCLM} + \mathcal{L}_{iou}, \quad (14)$$

where λ is the parameter used to balance MCLM loss and IoU loss, which is set to 1 by default for simplicity.

Table 3

Quantitative comparisons between the proposed MGCNet and 17 state-of-the-art methods. $\uparrow$ and $\downarrow$ mean that the higher and lower is better. The top three results are marked in red, blue and green, respectively.

Methods	Pub & Year	Cosal2015 [64]				CoSOD3k [65]				CoCA [41]			
		MAE $\downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$	MAE $\downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$	MAE $\downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$
GCAGC [39]	CVPR 2020	0.095	0.832	0.888	0.823	0.094	0.779	0.864	0.798	0.109	0.517	0.757	0.666
GICD [41]	CVPR 2020	0.071	0.844	0.887	0.844	0.079	0.770	0.848	0.797	0.126	0.513	0.715	0.658
ICNet [40]	NeurIPS 2020	0.058	0.858	0.900	0.857	0.089	0.762	0.845	0.794	0.147	0.513	0.704	0.654
CoADNet [44]	NeurIPS 2020	0.064	0.858	0.914	0.861	0.076	0.791	0.878	0.824	–	–	–	–
DeepACG [42]	CVPR 2021	0.064	0.842	0.892	0.854	0.089	0.756	0.838	0.792	0.102	0.552	0.771	0.688
CADC [46]	ICCV 2021	0.065	0.864	0.905	0.866	0.088	0.778	0.854	0.815	0.133	0.549	0.745	0.680
GCoNet [9]	CVPR 2021	0.068	0.847	0.887	0.845	0.071	0.777	0.860	0.802	0.105	0.544	0.760	0.673
CoEGNet [38]	TPAMI 2022	0.078	0.836	0.884	0.838	0.084	0.758	0.838	0.778	0.106	0.493	0.717	0.612
AIGA [66]	ICASSP 2022	0.080	0.812	0.865	0.829	0.088	0.746	0.836	0.788	0.148	0.466	0.681	0.628
DCFM [11]	CVPR 2022	0.067	0.856	0.892	0.838	0.067	0.805	0.874	0.810	0.085	0.598	0.783	0.710
HrSSMN [67]	TMM 2023	0.068	0.828	0.885	0.835	0.089	0.747	0.838	0.780	0.115	0.517	0.733	0.660
AGCNet [68]	TMM 2023	0.055	0.860	0.907	0.868	0.066	0.801	0.880	0.830	0.116	0.546	0.745	0.686
TCNet [48]	TCSVT 2023	0.054	0.867	0.912	0.873	0.068	0.804	0.878	0.832	0.101	0.552	0.777	0.685
CoRP [45]	TPAMI 2023	0.055	0.882	0.912	0.867	0.060	0.827	0.890	0.838	0.103	0.597	0.769	0.715
UFO [69]	TMM 2024	0.064	0.865	0.906	0.860	0.073	0.797	0.874	0.819	0.095	0.571	0.782	0.697
CADC++ [47]	TPAMI 2024	0.049	0.887	0.922	0.872	0.072	0.801	0.871	0.816	0.118	0.592	0.782	0.691
UniTR [70]	TMM 2024	0.065	0.857	0.906	0.860	0.072	0.802	0.879	0.827	0.096	0.567	0.782	0.703
MGCNet (ours)	–	0.048	0.887	0.923	0.877	0.057	0.827	0.897	0.837	0.092	0.600	0.786	0.708

**Fig. 6.** Comparisons with other methods in terms of PR and F-measure curves on three benchmark datasets.**Table 4**

Comparison with other methods on inference speed, computational complexity and parameter amount. The best results are in **bold**.

Model	GICD	CADC	CoADNet	TCNet	UFO	UniTR	Ours
Time (ms)	23.0	13.3	26.8	25.9	12.3	11.6	10.4
FLOPs (G)	46.8	91.8	120.4	40.7	218.4	67.6	34.4
Parameters (M)	278.0	329.9	29.7	69.3	69.6	72.3	32.2

4. Experiments

4.1. Datasets and evaluation metrics

To fully evaluate our proposed MGCNet, we conduct experiments on three widely used and challenging CoSOD datasets, namely, CoSal2015 [64], CoSOD3k [65], and CoCA [41]. CoSal2015 is the earliest published CoSOD dataset, comprising 50 groups with a total of 2015 images, each group containing one or more challenging issues such as variations in object size and pose, changes in background texture, and

diverse environments. CoSOD3k is the largest CoSOD dataset so far, which includes 3316 images divided in 160 groups with more high-quality pixel-level annotations. CoCA is the most challenging CoSOD dataset among these datasets, which consists of 1295 images divided into 80 groups. Compared to the other two datasets, CoCA contains more irrelevant salient objects in each image, which enables a more accurate evaluate the CoSOD method.

Following DCFM [11], we adopt the metrics of maximum F-measure (F_{β}^{max}) [71], mean absolute error (MAE) [72], S-measure (S_a) [73], and maximum E-measure (E_{ξ}^{max}) [74] to evaluate the performance of our model and other state-of-the-art models.

4.2. Implementation details

We use feature pyramid network [63] with VGG-16 [59] as our backbone. Following CoADNet [44], our network uses DUTS_class [41] and COCO-SEG [75] for training. Samples M are randomly selected from four different image groups as input:

$$M = \min(\#G_1, \#G_2, \#G_3, \#G_4, BS), \quad (15)$$

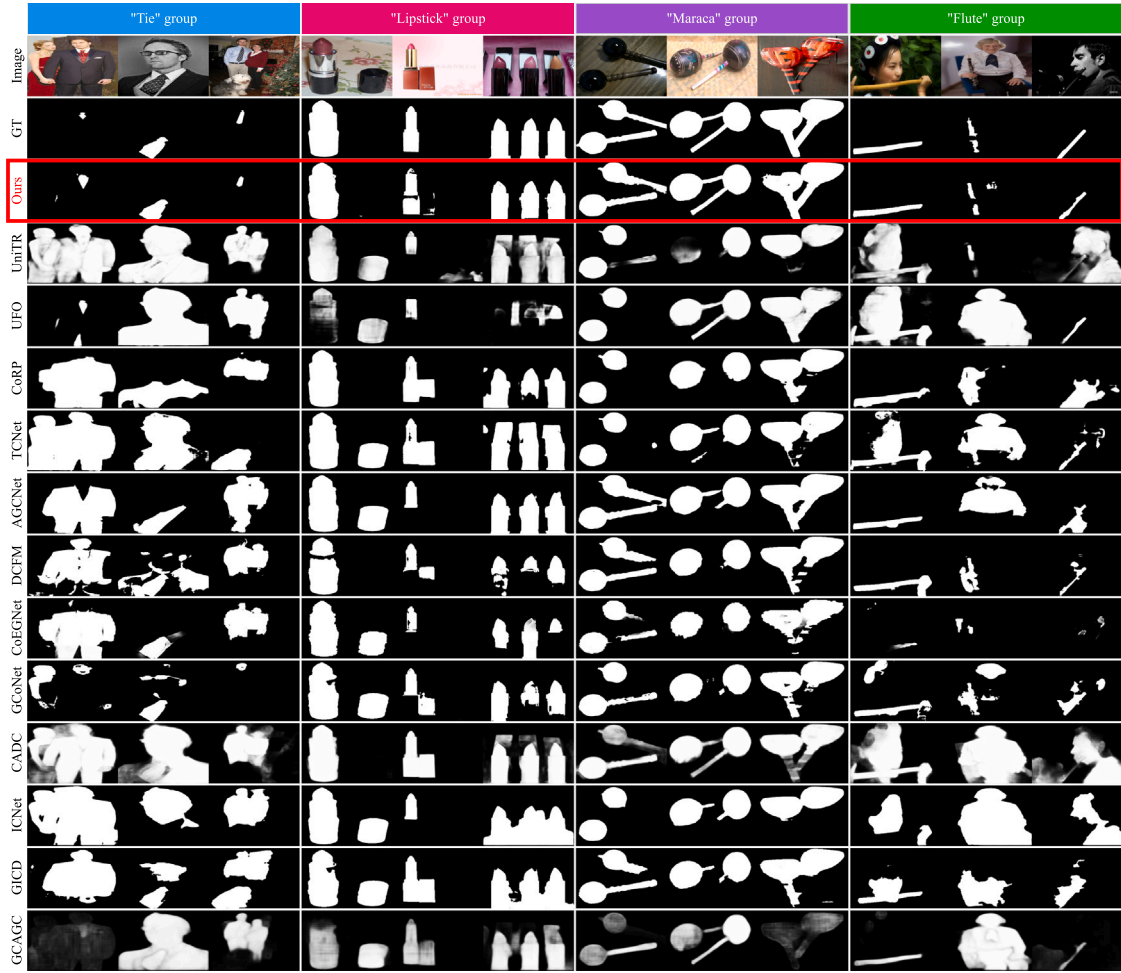


Fig. 7. Qualitative comparisons of our MGCNet and other methods.

Table 5

Ablation study of our proposed modules. Base means baseline. The best results are in **bold**.

ID	Base	Modules			Cosal2015				CoSOD3k				CoCA			
		MCLM	DEM	CRM	MAE ↓	$F_{\beta}^{max} \uparrow$	$F_{\xi}^{max} \uparrow$	$S_{\alpha} \uparrow$	MAE ↓	$F_{\beta}^{max} \uparrow$	$F_{\xi}^{max} \uparrow$	$S_{\alpha} \uparrow$	MAE ↓	$F_{\beta}^{max} \uparrow$	$F_{\xi}^{max} \uparrow$	$S_{\alpha} \uparrow$
1	✓				0.079	0.821	0.873	0.829	0.080	0.764	0.853	0.798	0.149	0.471	0.683	0.627
2	✓	✓			0.078	0.823	0.871	0.830	0.078	0.767	0.853	0.800	0.136	0.480	0.701	0.636
3	✓		✓		0.060	0.862	0.904	0.860	0.066	0.803	0.882	0.825	0.129	0.518	0.728	0.658
4	✓			✓	0.067	0.844	0.890	0.847	0.072	0.787	0.869	0.813	0.135	0.503	0.715	0.649
5	✓	✓	✓		0.053	0.876	0.915	0.868	0.059	0.820	0.894	0.835	0.099	0.590	0.772	0.701
6	✓	✓		✓	0.055	0.870	0.912	0.865	0.063	0.807	0.891	0.827	0.122	0.542	0.736	0.671
7	✓		✓	✓	0.053	0.875	0.916	0.868	0.058	0.825	0.896	0.836	0.098	0.590	0.781	0.701
8	✓	✓	✓	✓	0.048	0.887	0.923	0.877	0.057	0.827	0.897	0.837	0.092	0.600	0.786	0.708

where BS means the batch size that is set to 16 by default, M means the actual number of input images, and $\#$ denotes the image number in the corresponding group. The images are resized to 224×224 for both training and testing. To further enhance input data, three strategies of horizontal flipping, color enhancement, and rotation are applied. We follow UFO [69], using 4 transformer blocks with 4 multi-heads and setting the feature dimension of the transformer linear projection function to 782. Our MGCNet is trained for a total of 50 epochs with the AdamW optimizer [76]. The initial learning rate is set to $1e-4$ and weight decay is $1e-2$. The total training time of the model on a single NVIDIA GeForce RTX 3090 GPU is about 3.4 h.

4.3. Comparison with SOTAs

To comprehensively demonstrate the performance of our MGCNet, we compare the proposed approach with 17 deep learning-based

CoSOD methods, including GCAGC [39], GICD [41], ICNet [40], CoADNet [44], DeepACG [42], CADC [46], GCoNet [9], CoEGNet [38], AIGA [66], DCFM [11], HrSSMN [67], AGCNet [68], TCNet [48], CoRP [45], UFO [69], CADC++ [47] and UniTR [70]. Since CoSOD methods have developed considerably over the last few years and achieved better performance than single-SOD approaches, we do not list the single-SOD ones.

Quantitative Results: Table 3 shows the results of the quantitative comparison between our proposed MGCNet and previous competitive methods. As can be seen, the proposed method achieves the best overall results in almost all cases. Especially for the CoSal2015 dataset, our method is significantly superior to all competitors in terms of all metrics. For the CoSOD3k dataset, our method is generally ahead of all comparison methods, only slightly lagging behind the method CoRP in terms of S_{α} yet outperforming it equally or largely in terms of other metrics. For the CoCA dataset, our method surpasses all methods in

Table 6

Ablation study of different parts in MCLM. GL means only using global-level rather than local-level samples, while LL means only using local-level rather than global-level samples. The best results are in **bold**.

ID	Schemes	CoSal2015				CoSOD3k				CoCA			
		$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_{\alpha} \uparrow$	$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_{\alpha} \uparrow$	$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_{\alpha} \uparrow$
1	w/ GL	0.078	0.822	0.872	0.830	0.078	0.766	0.855	0.800	0.141	0.478	0.696	0.633
2	w/ LL	0.078	0.823	0.873	0.830	0.078	0.767	0.854	0.799	0.139	0.479	0.700	0.635
3	w/ GL and LL	0.078	0.823	0.871	0.830	0.078	0.767	0.853	0.800	0.136	0.480	0.701	0.636

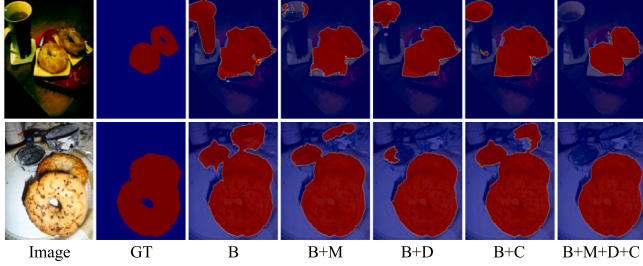


Fig. 8. Visual comparison of different proposed modules in ablation experiments. B, B+M, B+D, B+C and B+M+D+C denote baseline, baseline+MCLM, baseline+DEM, baseline+CRM and baseline+MCLM+DEM+CRM, respectively.

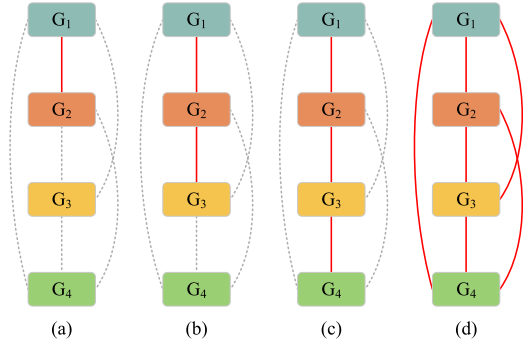


Fig. 9. Different connection manners of applying MCLM among four image groups. The solid and dashed lines represent explicit connection and disconnection, respectively. (a) Connection between the first two groups. (b) Connection between the adjacent two groups of the first three groups. (c) Connection between the adjacent two groups of all four groups. (d) Connection between any two groups of all four groups.

terms of the metrics of F_{β}^{max} and E_{ξ}^{max} , and achieves second and third place rankings in terms of the metrics of MAE and S_{α} , respectively. It is worth noting that the MAE of MGCNet falls behind DCFM on the CoCA dataset may be due to the fact that DCFM uses seed selection method to better suppress the complex background interference of images. In addition, we also compare our method with some advanced methods on inference speed, computational complexity and parameter amount in Table 4. Here, we set the batch size as 2 for all models during their inference. It can be seen that our method also has some advantages in terms of time, FLOPs and parameters. Meanwhile, we show the PR and F-measure curves for our method and the previous competitive methods in Fig. 6. As we can see, the PR curves generated by our method are generally better than those of other methods, especially on the CoSal2015 dataset. Also, the F-measure curves of our method are all significantly higher than the others, especially on the CoSal2015 and CoSOD3k datasets.

Qualitative Results: Fig. 7 shows a visual comparison of the proposed method with other compared methods, including four challenging image groups of tie, lipstick, maraca, and flute. It can be seen that most of the CoSOD methods for detecting co-salient objects in the tie and flute groups also detect people who are not co-salient objects. Although some methods such as AGCNet, CoRP, GCAGC, GICD and

Table 7

Ablation study of different connection strategies corresponding to four groups in MCLM. The best results are in **bold**.

ID	Strategies	CoCA			
		$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_{\alpha} \uparrow$
1	$n=1$	0.140	0.476	0.700	0.632
2	$n=2$	0.139	0.478	0.696	0.635
3	$n=3$	0.136	0.480	0.701	0.636
4	$n=6$	0.139	0.479	0.697	0.635

GCoNet are capable of excluding people in specific cases, their co-saliency maps do not accurately segment the boundaries of objects or contain a lot of background noise. Despite the fact that most of the methods are able to detect co-salient objects in both lipstick and maraca groups, they are all susceptible to noise and are unable to detect complete co-salient objects. In a word, compared to other methods, our method can exclude non co-salient objects while detecting co-salient objects more precisely, thus visually demonstrating the superiority of MGCNet.

4.4. Ablation study

We conduct ablation experiments to validate the effectiveness of each component (*i.e.*, CRM, DEM, and MCLM) embedded in our MGCNet. The quantitative results regarding each module and their combination are shown in Table 5. In addition, we also provide visualization results to demonstrate the effectiveness of proposed modules, shown in Fig. 8.

Effectiveness of MCLM: The multi-level contrastive learning module is employed to enhance the discriminative ability of the model to distinguish salient objects from other groups. Compared with the baseline model, the MCLM achieves the excellent performance improvement by 1.3%, 0.9%, 1.8% and 0.9% for the metrics of MAE , F_{β}^{max} , E_{ξ}^{max} and S_{α} on the CoCA dataset and a small improvement on the CoSal2015 and CoSOD3k datasets. It can be seen that, the MCLM shows great potential on CoCA dataset where images contain more interfering salient objects compared to the other datasets. In addition, the improvement is greater when MCLM is combined with the other two modules, which may be due to the fact that the model easily detects irrelevant salient objects after using the other modules, allowing MCLM to perform more effectively. To verify the effectiveness of global and local level samples in MCLM, we perform ablation studies on them in Table 6. Comparing with strategies that have only global-level samples or only local-level samples, our MCLM can take into account both global and local information to achieve the optimal performance. Furthermore, we explore different connection manners of applying MCLM among four image groups in Fig. 9. The four connection manners in the figure use 1, 2, 3 and 6 times MCLM, respectively. Particularly, since we are randomly selecting four image groups, we use (a) and (b) in the figure to represent the application of MCLM between two and three image groups. The (c) and (d) in the figure denote the different connection manners between the four image groups. The results of experiments using different numbers of MCLM are shown in Table 7, it can be found that our model has the most outstanding performance in applying MCLM between the adjacent two groups of all four groups, which also proves the superiority of our model using four image groups.

Table 8Ablation study of different parts in DEM. The best results are in **bold**.

ID	Schemes	Cosal2015				CoSOD3k				CoCA			
		$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$	$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$	$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$
1	w/ CFA	0.064	0.856	0.898	0.854	0.068	0.798	0.877	0.821	0.136	0.511	0.717	0.651
2	w/ SFA	0.064	0.856	0.899	0.854	0.068	0.798	0.878	0.820	0.139	0.509	0.714	0.650
3	w/ CFA and SFA	0.060	0.862	0.904	0.860	0.066	0.803	0.882	0.825	0.129	0.518	0.728	0.658

Table 9Ablation study of different parts in CRM. The best results are in **bold**.

ID	Schemes	Cosal2015				CoSOD3k				CoCA			
		$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$	$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$	$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$
1	w/ $\hat{F}_3$	0.074	0.828	0.879	0.836	0.078	0.771	0.859	0.802	0.144	0.488	0.696	0.636
2	w/ $\hat{F}_4$	0.068	0.845	0.890	0.847	0.074	0.783	0.866	0.812	0.143	0.499	0.705	0.645
3	w/ $\hat{F}_3$ and $\hat{F}_4$	0.067	0.844	0.890	0.847	0.072	0.787	0.869	0.813	0.135	0.503	0.715	0.649

Additionally, the performance of using MCLM between any two groups of all four groups decreases instead, where the reason may be that too many connections introduce additional interfering information.

Effectiveness of DEM: The dual enhancement module is adopted to mine the channel and spatial consensus information and enhance it to obtain more representative consensus features. Compared with the baseline model, the DEM achieves significant performance improvements on three datasets, especially on the CoCA dataset where the metrics of MAE , F_{β}^{max} , E_{ξ}^{max} and S_a are improved by 2.0%, 4.7%, 4.5% and 3.1%, respectively. Moreover, greater performance gains can be achieved when the DEM is used in combination with other modules. To verify the effectiveness of two components in DEM, we perform ablation studies on them in Table 8. As we can see, the proposed DEM achieves the most excellent performance when channel information and spatial information are extracted simultaneously. This indicates that both channel information and spatial information are important for extracting consensus features.

Effectiveness of CRM: The consensus retaining module is designed to retain the location information in the consensus features from being diluted. Compared with the baseline model, the CRM achieves satisfactory performance improvements on three datasets. For example, the performance improvements on the CoCA dataset are 1.4%, 3.2%, 3.2% and 2.2% respectively for the metrics of MAE , F_{β}^{max} , E_{ξ}^{max} and S_a . To verify the effectiveness of the different output branches in our proposed CRM, we perform ablation experiments in Table 9. Compared to the outputs of only branch $\hat{F}_3$ or only branch $\hat{F}_4$, the CRM fuses more detailed and semantic information to achieve the prominent performance, demonstrating that fusing features of neighboring layers from the encoder and adding them to the decoder is crucial for performance improvement.

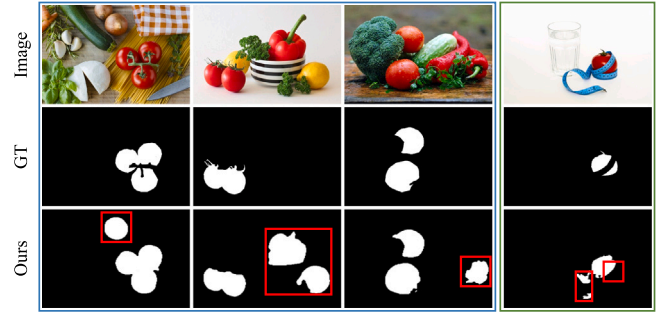
Effectiveness of Data Synthesis Strategy: Since CADG [46] proposes a data synthesis strategy to enrich the training dataset and has been applied to some methods such as DMT [49], we also explore the effectiveness of this strategy on our proposed method. The results are shown in Table 10. Comparing the first and second rows or the third and fourth rows in the table, the data synthesis strategy brings performance improvement in all four metrics. It can be seen that the data synthesis strategy can effectively improve the robustness and stability of the model.

4.5. Failure cases

We provide some failure cases of our approach in Fig. 10, where tomatoes are the co-salient objects in this group. There are two types of extreme scenarios that tend to constrain the detection performance of our model. For example, when there are objects in the image that are very similar or even identical in color or shape to the co-salient objects (e.g., the 1st to 3rd columns in the figure), our method will incorrectly detect them. This is due to the fact that our model is unable

Table 10Ablation study of data synthesis strategy. * indicates that we train the model using additional training dataset generated by data synthesis strategy. The best results are in **bold**.

ID	Models	CoCA			
		$MAE \downarrow$	$F_{\beta}^{max} \uparrow$	$E_{\xi}^{max} \uparrow$	$S_a \uparrow$
1	DMT	0.142	0.544	0.756	0.673
2	DMT*	0.123	0.599	0.786	0.706
3	MGCNet	0.092	0.600	0.786	0.708
4	MGCNet*	0.084	0.618	0.793	0.720

**Fig. 10.** Failures cases of our method.

to effectively distinguish the target from the background to some extent when it encounters such high similarity interference. In addition, when the co-salient objects are separated due to the occlusion of other objects (e.g., the 4th column in figure), it is also difficult for our method to detect all parts of the co-salient objects. This may happen because occlusion leads to feature loss or discontinuity, which partially weakens the global perception of the model. In such extremely challenging scenarios like this, the proposed method may not be successful in segmenting co-salient objects, which is also a common bottleneck in current methods.

5. Conclusion

In this paper, we propose a multiple group-wise correlation network termed MGCNet for co-salient object detection. Our network takes four image groups as inputs providing more positive and negative relations to obtain high-quality representations. We first introduce the multi-level contrastive learning module (MCLM), which construct more positive and negative sample pairs at global and local level to pull the images of the same group closer and push away the images of different groups. Afterwards, we design the dual enhancement module (DEM) to comprehensively extract intra-group consensus features in both channel dimension and spatial dimension via feature recombination and

multi-scale pooling fusion. Finally, we design the consensus retaining module (CRM) to supplement the location information lost by consensus features. We perform extensive comparison experiments using three challenging CoSOD datasets and four widely adopted evaluation metrics to show that the proposed MGCNet outperforms 17 previous SOTA methods in co-salient object detection, and also validate the effectiveness of the component modules through ablation experiments.

Although MGCNet has demonstrated satisfactory results in co-salient object detection, there is still room for improvement in the future. For example, exploring more effective strategies for feature fusion in the DEM could improve the accuracy of detecting co-salient objects. In addition, more lightweight and real-time models are also worth considering.

CRedit authorship contribution statement

Xian Fang: Writing – review & editing, Writing – original draft, Resources, Funding acquisition, Conceptualization. **Xin Wang:** Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jinchao Zhu:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization. **Qiaohong Chen:** Writing – review & editing. **Zuofan Chen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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